

Unit 3E: Numerical Surprises: Polygraphs, Mammograms, and More

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1. When and how individual trials may indicate a different result than the combined trials

This is about the danger of looking at small slices of data without considering the bigger picture. It's often called **Simpson's paradox**.

Example:

Imagine two colleges, College A and College B. Both have men and women applying to a graduate program.

- At College A:
 - 90 out of 100 men are admitted (90%).
 - 10 out of 20 women are admitted (50%).
- At College B:
 - 30 out of 100 men are admitted (30%).
 - 20 out of 40 women are admitted (50%).

If you look at each college separately:

- College A seems unfair to women (men 90% vs women 50%).
- College B seems fair (both around 30–50%).

But if you combine both colleges:

- Total men admitted = 120 out of 200 = 60%.
- Total women admitted = 30 out of 60 = 50%.

Now it looks like women are admitted less often overall, even though in **each college separately** they had similar or better chances.

Why does this happen? Because women may have applied more often to the harder college (with lower acceptance rates), while men applied more to the easier one. This flips the combined result.

Lesson: Looking at only individual parts can mislead you. Always check both the small groups and the combined data.

2. True positives, true negatives, false positives, and false negatives

Think of these as the **four possible outcomes of a test**.

- **True Positive (TP):** You are sick, and the test says you're sick. The test works correctly.
Example: You have malaria, and your malaria test comes back positive.
- **True Negative (TN):** You are healthy, and the test says you're healthy. Again, the test is correct.
Example: You don't have COVID, and your PCR test comes back negative.
- **False Positive (FP):** You are healthy, but the test wrongly says you are sick.
Example: A pregnancy test says positive, but you are not pregnant.
- **False Negative (FN):** You are sick, but the test wrongly says you are healthy.
Example: A COVID rapid test comes back negative, but you actually do have the virus.

Why is this important?

- False positives can cause **fear, stress, and unnecessary treatment**.
- False negatives can be even more dangerous because people may think they're safe and avoid treatment, spreading the disease.

3. When and how health tests, drug tests, and polygraphs can lead to surprising results

Health Tests:

Imagine a breast cancer screening test that is 95% accurate. Out of 10,000 women screened, maybe only 100 actually have cancer.

- The test correctly finds about 95 of them (true positives).
- But among the 9,900 women without cancer, about 5% (495 women) will still test positive falsely.

So, you get 95 true positives vs 495 false positives. That means most of the women who test positive don't actually have cancer. This is why doctors order further tests like biopsies.

Drug Tests:

Workplace drug tests sometimes give false positives. Eating poppy seed bread or taking certain medicines can trigger the test to say someone used drugs when they didn't. For the company, this is a serious issue because an innocent person may lose their job.

Polygraphs:

A polygraph measures stress signals like sweating or heartbeat. The problem is, people get nervous even when telling the truth. For example, an honest person under stress may fail, while a calm liar may pass. That's why polygraphs are not 100% trusted in courts.

Lesson: Tests can surprise us because accuracy depends not just on the test itself, but also on how common the condition is and how results are interpreted.

4. How a 90% accurate test may still yield more false positives than true positives

Let's use a step-by-step story.

Suppose there is a rare disease that affects 1 in 1,000 people. You take a test that is "90% accurate."

- Out of 1,000 people:
 - 1 person is truly sick. The test will likely detect them correctly (true positive).
 - 999 are healthy. With 90% accuracy, about 10% (100 people) will still test positive by mistake (false positives).

So when you get the results:

- Positive results = 1 true positive + 100 false positives = 101 positives.
- Only 1 of those 101 is actually sick.

That means if you test positive, the chance you really have the disease is only about **1 in 101 (less than 1%)**, even though the test is "90% accurate."

Lesson: A test's accuracy doesn't guarantee reliable results when the condition is rare. This is called the **base rate problem** in statistics.

5. How Democrats and Republicans can make different claims about the same data

This is about **framing**. The numbers are the same, but each side chooses what to highlight.

Example: Economy and Jobs

- Suppose unemployment drops from 10% to 6%.
- Democrats: "Unemployment has dropped by 4%, millions of jobs created."
- Republicans: "Unemployment is still very high at 6%, meaning millions are still jobless."

Both are correct, but the **emphasis is different**.

Example: Government Spending

- Democrats might say: "We increased education spending by 20%."
- Republicans might say: "The budget deficit also grew by billions."

Again, both are true. Each side highlights what supports their argument.

Lesson: Data itself doesn't lie, but the way it is presented can lead to different interpretations. As a critical thinker, you should always ask: *What part of the data is being left out?*